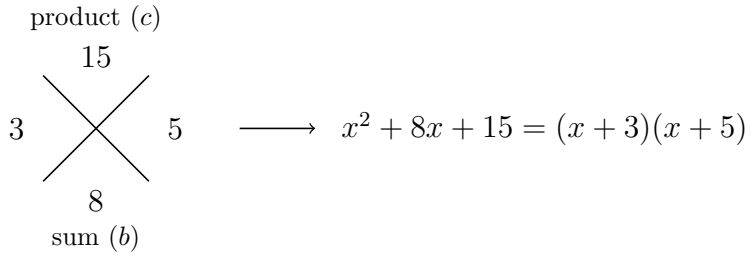
Factoring Quadratics & the Zero Product Property

Name: _____ Date: _____

Example: The X Method

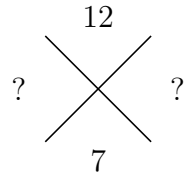
To factor $x^2 + 8x + 15$, find two numbers whose *product* is $c = 15$ (top) and whose *sum* is $b = 8$ (bottom). The pair goes on the sides.

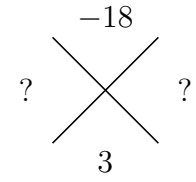


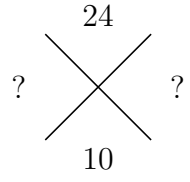
Since $3 \times 5 = 15$ and $3 + 5 = 8$, the numbers are 3 and 5, so the factors are $(x + 3)$ and $(x + 5)$.

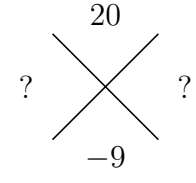
A. Warm-Up: The Diamond (X) Method

Fill in the sides of each X: the two numbers must multiply to the top and add to the bottom.

1. 

3. 

2. 

4. 

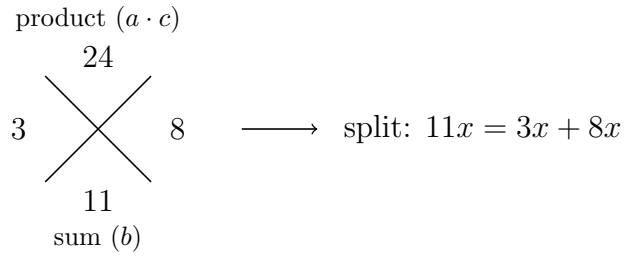
B. Factor Each Trinomial

Use the X method or an area (box) model. Check by multiplying.

1. $x^2 + 7x + 12$
2. $x^2 + 9x + 18$
3. $x^2 + 11x + 24$
4. $x^2 - x - 20$
5. $x^2 - 10x + 21$
6. $x^2 + 2x - 35$
7. $x^2 - 13x + 40$
8. $x^2 - 4x - 45$

Example: Factor by Grouping

When $a \neq 1$, as in $2x^2+11x+12$, the X still works — but the *product* is now $a \cdot c = 2 \cdot 12 = 24$, and the pair splits the middle term instead of giving the factors directly.



$$
\begin{aligned}
2x^2 + 11x + 12 &= 2x^2 + 3x + 8x + 12 && \text{split the middle term} \\
&= (2x^2 + 3x) + (8x + 12) && \text{group} \\
&= x(2x + 3) + 4(2x + 3) && \text{factor each group} \\
&= (2x + 3)(x + 4) && \text{common factor } (2x + 3)
\end{aligned}
$$

C. Factor by Grouping

Split the middle term, then group. *Hint: find two numbers whose product is $a \cdot c$ and whose sum is b.*

1. $2x^2 + 7x + 3$

2. $3x^2 + 10x + 8$

3. $4x^2 - 4x - 15$

4. $6x^2 + 11x - 10$

D. Solve Using the Zero Product Property

Get one side equal to 0, factor, then set each factor equal to 0.

1. $(x+2)(x-7)=0$

2. $x^2+8x+15=0$

3. $x^2-5x-14=0$

4. $x^2+6x=27$

5. $2x^2+5x-3=0$

6. $x^2=9x$ *(Careful: don't divide both sides by x!)*

Answer Key

A. Warm-Up

1. 3 and 4
2. 4 and 6
3. 6 and -3
4. -4 and -5

B. Factoring

1. $(x+3)(x+4)$
2. $(x+3)(x+6)$
3. $(x+3)(x+8)$
4. $(x-5)(x+4)$
5. $(x-3)(x-7)$
6. $(x+7)(x-5)$
7. $(x-5)(x-8)$
8. $(x-9)(x+5)$

C. Grouping

1. $2x^2+x+6x+3 = x(2x+1)+3(2x+1) = (2x+1)(x+3)$
2. $3x^2+6x+4x+8 = 3x(x+2)+4(x+2) = (3x+4)(x+2)$
3. $4x^2+6x-10x-15 = 2x(2x+3)-5(2x+3) = (2x+3)(2x-5)$
4. $6x^2+15x-4x-10 = 3x(2x+5)-2(2x+5) = (3x-2)(2x+5)$

D. Zero Product Property

1. $x=-2$ or $x=7$
2. $(x+3)(x+5)=0$, so $x=-3$ or $x=-5$
3. $(x-7)(x+2)=0$, so $x=7$ or $x=-2$
4. $x^2+6x-27=0$; $(x+9)(x-3)=0$, so $x=-9$ or $x=3$
5. $(2x-1)(x+3)=0$, so $x=\frac{1}{2}$ or $x=-3$
6. $x^2-9x=0$; $x(x-9)=0$, so $x=0$ or $x=9$ (dividing by x loses the solution $x=0$)